

Technical Project Overview: Supply Chain Sentinel

1. Project Description

The Supply Chain Sentinel ecosystem is an automated, cloud-based framework built using the Microsoft Power Platform. It bridges conversational AI with background enterprise automated loops to track parts logistics, automatically assess alternative vendor data, request human approval over secure channels, and instantly send out high-priority external sourcing emails within minutes of a component shortage detection.

2. Structural Architecture

The demonstration video shows four distinct modules functioning seamlessly in tandem:

- **Microsoft Copilot Studio Agent:** A conversational layer named 'Supply Chain Sentinel' built to receive inventory anomalies and route automated alerts. (Note: Environment is maintained in a verified 'Draft' testing state during backend cloud provisioning transitions).
- **Microsoft Lists (Disruption Manifest):** A clean data framework acting as the central organizational ledger for logging shortages, burn rates, and vendor assignments.
- **Core Mitigation Flow (Power Automate):** An event-driven cloud flow triggered by new entries in the manifest list. It requests pricing metrics via HTTP, writes live updates to the ledger, and handles real-time adaptive cards.
- **Governance Execution Flow (Power Automate):** A secondary conditional background compliance flow that triggers via updates to checking statuses, providing automated external procurement communications.

3. Step-by-Step Video Walkthrough Guide

Reviewers may use this sequential guide to interpret the on-screen events of the silent demonstration video:

- **00:00 - 00:15 | Environment Overview:** Opening of Microsoft Copilot Studio displaying the customized conversational 'Supply Chain Sentinel' structural agent workspace.
- **00:15 - 00:33 | Logic & Database Review:** Inspection of the agent's core step-by-step designer triggers followed by a shift to the initially empty 'Disruption Manifest' workspace inside Microsoft Lists.
- **00:33 - 01:10 | Power Automate Frameworks:** Review of the mitigation and governance logic trees before executing a manual system initialization test.
- **01:10 - 01:35 | Simulating an Anomaly:** The user manually inputs a critical logistics event into the manifest: Part Deficit SKU-TURBINE-2026, totaling a shortage of 1,200 units.
- **01:35 - 02:24 | Automated Triage Run:** Power Automate catches the trigger event immediately, gathers active marketplace data via an HTTP request layer, and safely transitions to a pending approval state.
- **02:24 - 03:45 | One-Click Executive Sign-off:** A secure card notification surfaces in Microsoft Teams. The operations manager analyzes the calculated parameters and clicks 'Authorize Copilot Sourcing Engine' directly within the workspace.
- **03:45 - 04:58 | Procurement Dispatch:** The manager's authorization updates the internal manifest ledger data parameters and commands the secondary governance system to automatically email external suppliers an urgent, 4-hour quote request.
- **04:58 - 06:33 | Resolution and Audit Logs:** The database systematically displays an automated alternative vendor allocation value update of \$200,000.00 with an approved status tag, wrapping up with a clean verification of error-free system activity run histories.